

# Nidhi Isotiya (she/her)

Pursuing MS in Computer Science, Seattle, WA

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## EDUCATION

**2023 - 2025 M.S. – Computer Science, Seattle University**

**2016 - 2020 B.E. – Environmental Engineering, Marwadi Education Foundation**

## PROJECTS

### Scalable Image Processing and Hosting Pipeline (Java, AWS)

- Developed a cloud-native system for high-volume user image uploads using S3 presigned URLs, storing metadata and job state in Postgres for reliable tracking.
- Implemented Lambda to validate uploads and enqueue jobs in SQS, while workers on ECS and Fargate process images (resize, compress) and uploads them to S3.
- Delivered processed images globally via CloudFront, demonstrating a scalable, event-driven architecture that integrates serverless triggers with relational databases.

### Tutorial Application (TypeScript, MongoDB, AWS)

- Built a full-stack web application allowing users to upload and share repair tutorials; implemented features for browsing, keyword search, likes/dislikes, and comments to enhance user engagement.
- Developed backend services using Node.js, TypeScript, and Mongoose, with data persistence in MongoDB; integrated Google OAuth for secure authentication and streamlined user onboarding.
- Deployed and containerized application on AWS EC2 with Docker; configured Auto Scaling Groups and Application Load Balancer to ensure high availability and scalability.

### eCommerce Platform (React, Node.js, MongoDB)

- Developed a full-stack eCommerce application featuring shopping cart, product reviews, pagination, admin product management, and user profiles with order history.
- Implemented responsive UI using React functional components, hooks, React Router, and React-Bootstrap; managed global state efficiently with Redux.
- Designed and built RESTful APIs with Express and Node.js; leveraged MongoDB with Mongoose for schema modeling and implemented secure cookie-based authentication.

### Stock Price Prediction using RNNs (AI)

- Built and trained Recurrent Neural Network models (Simple RNN, GRU, LSTM) to predict Microsoft (MSFT) stock prices and trading volume.
- Preprocessed historical stock data with time-series windowing and MinMax scaling, shaping inputs for sequential learning. Predicted Open, High, Low, Close, and Volume for a 2-week horizon
- Evaluated models using MSE/MAE, compared performance across architectures, and visualized predicted vs. actual trends.

### Apartment Connect Application (Flutter, Firebase)

- Built an Android mobile application in Flutter to streamline communication between apartment residents and management.
- Designed resident features including secure authentication, maintenance requests, announcements, message boards, real-time chat, and profile management with customizable notifications.
- Implemented management features such as user administration, posting announcements, and resolving complaints and maintenance requests through an integrated dashboard.
- Leveraged Firebase Authentication, Firestore Database, Storage, and Cloud Messaging to provide real-time updates and reliable backend services.

## TECHNICAL SKILLS

**Languages:** Java, C++, C#, Python, TypeScript, SQL, HTML, CSS, JavaScript

**Frameworks:** Kubernetes, React, Flutter, Node.js, Tensorflow, JUnit, RESTful API

**Developer Tools:** AWS, Kubernetes, Docker, Firebase, Git, VS Code

**Databases and Messaging:** MySQL, PostgreSQL, MongoDB, Redis, Kafka